One-page summary: a pass selector keeps ZX rewriting only where it lowers the cost of a circuit on Quantinuum H2

Team Learned ZX Rewriting: C.-F. Kam, A. Mahasinghe, D. De Silva, N. Karunarathne, F. Cadet, M. Bessafi, K. De Silva.
Quantinuum Singapore Grand Challenge, submission 1. Main report `report.pdf`, technical appendix `report_full.pdf` (cited as "App."), source `code.zip` with `README.md`, emulator evidence `EVIDENCE.md`. `[F]` marks a checkable statement, `[O]` a judgement.

Problem statement

`[F]` A run on Quantinuum H2 is billed in HQC = 5 + C (N_1q + 10 N_2q + 5 N_SPAM) / 5000 (App. 1.2). ZX rewriting (PyZX teleport_reduce) placed before pytket's Quantinuum pass can lower this cost or raise it. Always run, it costs 6.27 % more HQC than pytket alone on 180 Trotter circuits and 22.1 % more on 3 QFT adders. pytket's own ZX pass `ZXGraphlikeOptimisation` is documented as one that "may increase the cost of the circuit". `[O]` A team with a fixed HQC budget cannot tell in advance, circuit by circuit, whether ZX rewriting pays.

Approach

`[F]` The pass selector PSEL (`pipelines/pass_select.py`) compiles each circuit twice: with P0, pytket-quantinuum's default pass at optimisation level 2, and with P2, PyZX teleport_reduce followed by P0. It drops a candidate that fails its exactness check and keeps the one with the lower HQC. Our P0 and P2 give the same counts as the official pytket-quantinuum pass on all 445 compiled rows of families A, B1, B1x and B2 (App. 2.2). `[O]` PSEL is pytket's `PassSelector` pattern with HQC as the metric; the contribution is the measured cost ledger and the guard result, not a new selection mechanism. `[F]` On the 20 circuits of family C (20 to 56 qubits) a PyZX miter reduction proves the PSEL output equal to its input on 18, three of them after snapping angles by at most 5.27e-12 half-turns (App. 3.8). The team's exactness-barrier certificate (arXiv:2608.22801) gives a per-instance floor on the T-count of any exact rewrite.

Key results

- `[F]` 15 arithmetic and oracle circuits of 20 to 56 qubits without continuous rotation angles (9 adders, 6 multi-controlled X oracles): PSEL saves 748.4 of 10,019.6 HQC or 7.47 % against P0, and 12.9 % on the adders. The PSEL output of all 15 is proven equal to its input as a full unitary up to global phase. Family C is count-only apart from one H2-2E run. Over all 20 circuits of family C PSEL saves 4104.4 HQC or 16.7 %; two multipliers with continuous angles carry 81.8 % of that saving.
- `[F]` Guard: against always running P2, PSEL avoids 1,076.4 HQC on the 180 Trotter circuits and 1,057.2 HQC on the 3 QFT adders. PSEL costs no more HQC than P0 whenever P0 passes its checks, by construction.
- `[F]` Clifford+T benchmarks B1 (15 circuits, n <= 12): 743.6 HQC or 4.79 % saved, median per circuit 3.94 %; hwb8 alone gives 84.5 % of that saving.
- `[F]` Worked use case, Grover search on 7 and 9 qubits: PSEL saves 5.42 % and 4.96 % of the HQC and affords 791 and 322 more shots within a 10,000 HQC budget.
- `[F]` Magic states: the ZX stage removes 40.6 % of the written T-count on B1. On 20 synthesised Haar circuits of 2 and 3 qubits the certified floor is 0.997 of the T-count left after ZX, pooled: further exact T optimisation can gain at most 0.3 % pooled and 0.72 % on the worst instance.
- `[F]` Emulator execution: 20 H2-2E jobs (6598.0 HQC by the cost formula) and 3 H2-Emulator jobs, job IDs in `EVIDENCE.md`. The 22-qubit adder vbe_adder_7 ran under P0 and P2 (F_H 0.886 and 0.897, z = 0.79 at 1000 shots). At 4000 shots per pipeline the Hellinger fidelity of P2 minus P0 on vbe_adder_3 is +0.0108 (z = +2.38, p = 0.017; Holm-adjusted 0.052 over three circuits); grover_5 shows no gain. The fidelity gain is therefore not established at 0.05 after correction. Nothing has run on hardware.

Relevance to quantum computing

`[F]` On family C PSEL saves most on adders and multipliers. The cited literature places adders in the modular arithmetic of Shor factoring (App. 1.4) and arithmetic in the path loading and payoff of derivative pricing (App. 1.5; Chakrabarti et al., arXiv:2012.03819). On a surface-code machine each T gate consumes one magic state (Litinski, arXiv:1808.02892). `[O]` PSEL lowers the cost of these subroutines on H2 today. The certificate tells a fault-tolerant compiler team per instance that exact rewriting of synthesised rotations is not where its effort pays.

Potential impact

`[O]` At the Azure Quantum plan averages of USD 12.50 and 10.29 per HQC (App. 1.5) the 748.4 HQC saved on family C is worth about USD 7,700 to 9,400, and the 1,076.4 HQC avoided on the Trotter circuits about USD 11,100 to 13,500, each circuit counted once at 1000 shots. PSEL drops into a pytket workflow for the price of a second compilation. `[F]` Limits: no user interview or survey is part of this submission; family C is count-only apart from one emulator run; no hardware run. `[O]` Finals plan: 4364 shots per pipeline on vbe_adder_3 at 1147.3 HQC, power 0.80 if the emulator gain holds on hardware (App. 6).